PONY ENV

RP2350 Hardware Reference Document

Rev 0.1 — DRAFT

1. Overview

Pony ENV is a compact Eurorack envelope generator with a built-in stereo VCA. The module is built around the Raspberry Pi RP2354B microcontroller and uses a PCM5100 stereo I2S DAC for CV output and an AS3360 dual VCA for audio processing.

The envelope offers independent Attack and Decay time controls with CV inputs, four selectable envelope shapes, four time range multipliers, and four operating modes including one-shot, ASR, loop, and ping. A NeoPixel LED on the panel reflects envelope activity with brightness proportional to the envelope level. The envelope can be triggered via the front panel trigger jack or the TAP button.

2. Key Hardware Components

2.1 Microcontroller — RP2354B

The RP2354B is the B-package variant of the RP2350, featuring extended ADC-capable GPIOs up to GPIO 47. This allows direct ADC connection of the three potentiometers at GPIOs 45, 46, and 47 without additional multiplexing.

2.2 DAC — PCM5100 (I2S)

The PCM5100 is a stereo I2S DAC used to generate the two envelope CV outputs:

- L channel — drives the AS3360 VCA amplitude (both channels) and is also available at the Env Atte jack.
- R channel — raw envelope output available at the Env Out jack.

> **i** *I2S connected to GPIO 0 (BCLK), GPIO 1 (LRCLK), GPIO 2 (DIN).*

2.3 Dual VCA — AS3360

The AS3360 is a classic dual VCA integrated circuit. Both VCA channels share a common CV derived from the PCM5100 L output. Audio input conditioning for the AS3360 is handled externally and is outside the scope of this document.

2.4 CV Input Conditioning — MCP6002

Each CV input (Attack, Decay, LVL CV) is summed with its corresponding potentiometer value using an MCP6002 dual op-amp configured as a summer. The MCP6002 clips the output if the sum exceeds 10V, preventing over-range. The summed signal is then scaled to 0–3.3V for the RP2350 ADC inputs.

2.5 Trigger Interface

Both trigger input and EOC output use NPN transistor interfaces:

- Trigger In (GPIO 12) — NPN converts incoming Eurorack gate to 3.3V logic for the RP2350.

- EOC Out (GPIO 18) — NPN drives +12V Eurorack gate output when GPIO 18 is set HIGH.

3. GPIO Pin Assignment

GPIO	Function	Direction	Notes
0	I2S BCLK	OUTPUT	PCM5100 bit clock
1	I2S LRCLK	OUTPUT	PCM5100 left/right word clock
2	I2S DIN	OUTPUT	PCM5100 data in
12	Trigger In	INPUT	NPN transistor interface — RP2350 sees 3.3V logic
18	EOC Out	OUTPUT	NPN transistor drives +12V Eurorack gate when HIGH
19	Range selector bit 0	INPUT	4-position selector, internal pull-up, active low
33	Mode selector bit 3	INPUT	4-position selector, internal pull-up, active low
34	Shape selector bit 1	INPUT	4-position selector, internal pull-up, active low
35	Shape selector bit 0	INPUT	4-position selector, internal pull-up, active low
36	Range selector bit 3	INPUT	4-position selector, internal pull-up, active low
37	Shape selector bit 2	INPUT	4-position selector, internal pull-up, active low (note: non-sequential)
38	Range selector bit 2	INPUT	4-position selector, internal pull-up, active low
39	Range selector bit 1	INPUT	4-position selector, internal pull-up, active low
40	Mode selector bit 2	INPUT	4-position selector, internal pull-up, active low
41	Mode selector bit 1	INPUT	4-position selector, internal pull-up, active low
42	Mode selector bit 0	INPUT	4-position selector, internal pull-up, active low
43	NeoPixel LED	OUTPUT	WS2812 single pixel — envelope activity indicator
44	Button	INPUT	Internal pull-up, active low — manual trigger + TAP
45	LVL pot + LVL CV	ADC INPUT	0–3.3V after MCP6002 summer (pot + 0–10V CV input)
46	Decay pot + CV	ADC INPUT	0–3.3V after MCP6002 summer (pot + 0–10V CV input)
47	Attack pot + CV	ADC INPUT	0–3.3V after MCP6002 summer (pot + 0–10V CV jack)

i *Shape selector bit mapping uses non-sequential GPIOs (35, 34, 31, 37). Firmware must account for this ordering.*

4. Jack Connections

Jack	Direction	Description
Trigger In	INPUT	Eurorack gate/trigger input. NPN transistor converts to 3.3V for RP2350. Fires envelope on rising edge (or holds in ASR mode).
LVL CV	INPUT	0–10V CV input summed with LVL pot via MCP6002 op-amp. Controls envelope output amplitude.
Audio In L	INPUT	Left audio channel input to AS3360 dual VCA. Level conditioned externally.

Jack	Direction	Description
Audio In R	INPUT	Right audio channel input to AS3360 dual VCA. Level conditioned externally.
Env Out	OUTPUT	Raw envelope CV output. Driven by PCM5100 R channel.
Env Atte Out	OUTPUT	Attenuated envelope CV output. Driven by PCM5100 L channel. Also controls AS3360 VCA amplitude for both audio channels.
Audio Out L	OUTPUT	Left audio channel output from AS3360 dual VCA.
Audio Out R	OUTPUT	Right audio channel output from AS3360 dual VCA.
EOC	OUTPUT	End of Cycle gate output. NPN transistor drives +12V Eurorack gate when GPIO 18 is HIGH. Fires at end of each envelope cycle.

5. CV Input Specifications

Parameter	Jack Range	ADC Range	Notes
Attack	0–10V	0–3.3V	Summed with Attack pot (GPIO 45). MCP6002 clips sum at 10V before scaling.
Decay	0–10V	0–3.3V	Summed with Decay pot (GPIO 46). MCP6002 clips sum at 10V before scaling.
LVL CV	0–10V	0–3.3V	Summed with LVL pot (GPIO 47). Controls envelope output amplitude.

ℹ *All CV inputs expect 0–10V Eurorack standard. The MCP6002 summer clips at 10V and scales to 0–3.3V before the ADC. Negative CV voltages are not supported.*

6. Selector Switch Logic

Each 4-position selector has four pins connected to individual GPIOs. All selector inputs use internal pull-ups and are active low. At any given time, exactly one pin will read LOW (the selected position); the others will read HIGH. The firmware reads the four bits and maps them to the corresponding function.

6.1 Envelope Mode Selector

GPIO order: [42, 41, 40, 33]

Position	Bits [42, 41, 40, 33]	Mode
1	1000	One-shot — triggers once, completes A+D, then idles
2	0100	One-shot ASR — Attack, Sustain while trigger held, Release on trigger off
3	0010	Loop — continuously cycles A+D while active
4	0001	Ping — alternates forward and reverse on each trigger

6.2 Envelope Shape Selector

GPIO order: [35, 34, 31, 37] — Note: non-sequential GPIO numbering.

Position	Bits [35, 34, 31, 37]	Shape
1	1000	Linear — constant rate of change throughout segment
2	0100	Exponential — fast initial rise/fall, slowing toward target
3	0010	Logarithmic — slow initial rise/fall, accelerating toward target
4	0001	Exp-Log — exponential attack, logarithmic decay (to be confirmed with developer)

ℹ *Exact curve parameters and crossover points for the Exp-Log shape are to be agreed with the developer. The AdEnv module from DaisySP provides a curve scalar parameter that can implement these shapes.*

6.3 Time Range Selector

GPIO order: [19, 39, 38, 36]

Position	Bits [19, 39, 38, 36]	Range
1	1000	Fast — short time constants (exact values TBD)
2	0100	Medium — moderate time constants (exact values TBD)
3	0010	Slow — long time constants (exact values TBD)
4	0001	Extended — very long time constants (exact values TBD)

ℹ *Exact time constant values for each range are to be determined during firmware tuning. Range acts as a multiplier applied to the Attack and Decay pot values.*

7. Button and LED

7.1 TAP Button (GPIO 44)

The button uses the RP2350 internal pull-up and reads active low. It serves two functions:

- Manual trigger — pressing fires the envelope identically to a trigger jack input.
- TAP — function TBD (reserved for future use such as tap tempo).

7.2 NeoPixel LED (GPIO 43)

A single WS2812 NeoPixel is connected to GPIO 43. It displays envelope activity: brightness tracks the current envelope level in real time. At envelope peak the LED is at full brightness; at idle it is off. Color and behavior per mode can be defined in firmware.

8. Audio Signal Routing

PCM5100 L output	→	AS3360 VCA CV (both channels) + Env Atte jack
PCM5100 R output	→	Env Out jack (raw envelope CV)
Audio In L + R	→	AS3360 dual VCA inputs
AS3360 VCA outputs	→	Audio Out L + R jacks

9. Firmware Notes

9.1 DSP Library

The Pifosti codebase (DaisySP, MIT licensed) provides ready-to-use envelope generators:

- AdEnv — Attack/Decay trigger-based envelope. SetCurve() parameter supports linear, exponential, and logarithmic shapes. Best suited for one-shot and loop modes.
- Adsr — Full ADSR gate-based envelope. Best suited for one-shot ASR mode where the gate holds the sustain.

9.2 NeoPixel

The WS2812 on GPIO 43 is driven via PIO (same approach as the Pifosti ws2812 example). Brightness is set proportionally to the current envelope output value each sample period.

9.3 Selector Reading

Each 4-position selector is read by scanning its four GPIOs and finding the active-low pin. The non-sequential GPIO mapping of the Shape selector ([35, 34, 31, 37]) must be handled explicitly in firmware — do not assume bit position corresponds to GPIO order.

9.4 Open Items

1. Exact time constant values for Fast / Medium / Slow / Extended ranges — to be determined during tuning.
2. Envelope shape curve parameters (especially Exp-Log crossover) — to be agreed with developer.
3. Ping mode direction logic and retrigger behaviour — to be defined.
4. TAP button secondary function — to be defined.
5. Trigger input GPIO pin — not yet confirmed (GPIO 12 per current spec).